SHRIPAD DESHMUKH

MEDIA & DATA SCIENCE RESEARCH LAB, ADOBE, INDIA

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Research Background	Reinforcement Learning	*Interpretability of RL policies, Safe RL, Inverse RL, Multi-Agent Collaboration*
	Document Structure Extraction	*Multi-modal Element Association, Object Detection, Semantic Segmentation, Graph Convolutional Networks, Vision Transformers*

Education

Indian Institute of Technology, Madras ***Chennai, TN***
B.Tech. in Electrical Engineering, CGPA: **9.20/10** *Aug 2016 – Jul 2020*

Publication

Nikaash Puri, Sukriti Verma, Piyush Gupta, Dhruv Kayastha, **Shripad Deshmukh**, Balaji Krishnamurthy, Sameer Singh. "Explain Your Move: Understanding Agent Actions Using Focused Feature Saliency". *International Conference on Learning Representations* (**ICLR**), 2020.

Patents

- **Shripad Deshmukh**, Milan Aggarwal, Mausoom Sarkar, Hiresh Gupta. "Refining Element Associations for Form Structure Extraction". *Attorney Docket No. P10768-US, 2021. (filed)*
- Sukriti Verma, **Shripad Deshmukh**, Jayakumar Subramanian, Piyush Gupta, Nikaash Puri. "Customer Journey Management Using Machine Learning". *Attorney Docket No. P10405-US*, 2021. *(filed)*

Work Experience

Media & Data Science Research Lab, Adobe ***Noida, UP***

Machine Learning Engineer, *Guide: Balaji Krishnamurthy* *Jun 2021 – Present*

Benchmarking Structure Extraction Models *Sept 2021 – Nov 2021*

- Initiated conversion of the proprietary **forms dataset** into **standard COCO[1] format** for continuous evaluation of Adobe's AFCS[2] production models against upcoming leaders in the domain
- Assessed & documented **5** top **semantic segmentation** and **object detection** approaches in terms of their accuracy, time taken for conversion of an individual PDF form, no. of post-conversion user corrections
- Achieved average boost of **7**% in *conversion quality*[3] over all the classes & **deployed 3** models

Table Detection *Jun 2021 – Sept 2021*

- Trained & analyzed **4** different object detection-based models for **detecting bounds of tables** in documents
- Boosted the conversion quality for class '*Table*' by more than **8.5%** and **deployed** the model to production

Forms Correction Propagation *May 2021 – Aug 2021*

- Designed a novel paradigm to solve the problem of **propagating user-specific changes** in the extracted structure of one form **to other similar regions** in the entire form corpus
- The algorithm provides high **customizability**, option of **undoing, ease of deployment** and total **protection against changes to original weights** of the extraction model
- Demonstrated **Proof-of-Concept** to **Adobe Document Cloud**, working on publishing & patenting the same

Member of Technical Staff, *Guide: Balaji Krishnamurthy* *Aug 2020 – May 2021*

Multimodal Element Association *Oct 2020 – Feb 2021*

- Debugged **TF1.x** codebase for inclusion of segmentation priors in multimodal association network, achieving **15%** precision & **4%** recall improvement in extraction of higher-order structures; **deployed** it to AFCS
- **Drafted** patent-idea for patent '*Refining Element Associations …*'; listed as **exemplary drafts** on Adobe Brightidea site (**only new addition** to the list in year 2021 till September; **1 in 300+** applications)

Customer Journey Management *Oct 2020 – Dec 2020*

- Ideated **use of trajectory based IRL[4]** methods for **reducing cost of intervention** during customer journeys
- Simulated the journeys & achieved **more than 20% reduction** in churn rate in batch learning settings

Research Intern, *Mentors: Nikaash Puri, Sukriti Verma, Piyush Gupta* *May 2019 – Aug 2019*

Explaining Reinforcement Learning Agents

- Collaboratively proposed a new **attribution metric for visual explanations** of RL policies, significantly outperforming prior art in terms of **specificity & relevance** of generated saliency maps
- Co-produced explanations for **Stockfish** chess engine; extended the work to **Go & Atari** games; published the research in **ICLR 2020**
- Additionally, explored the direction of using **entropy-based techniques** to explain the RL agent's actions
- Received **Pre-Placement Offer** to work at Media & Data Science Research Lab as **a full-time** employee

[1]*Common Objects in Context* [2]*Automated Forms Conversion Service* [3]*Captures post-conversion corrections* [4]*Inverse Reinforcement Learning*

Center for Development of Advanced Computing (CDAC) ***Pune, MH***

R&D Intern, *Mentors: Dr. Narendra Karmarkar & Dr. VCV Rao* *Dec 2018 – Jan 2019*

Matrix and Graph Computations in High Performance Computing, Big-Data Environment

- Part of **NSM**[5], a project aimed at creating **a cluster of 70+ supercomputers** to empower Indian R&D sector
- Developed FORTRAN libraries to test & extract performance of **Intel NLA**[6] **Kernels** for different matrix types
- Studied algorithms for generating **hypergraphs** for given no. of nodes, edge probabilities, degree of nodes.
- Integrated all existing modules under **Unified Object-Oriented Interface** along with the documentation

BigCat Wireless, IITM-Research Park ***Chennai, TN***

Communication Systems Intern, *Mentor: Kannan G* *May 2018 – July 2018*

Self-Interference Cancellation in Full Duplex 5G communication Systems

- Corrected self-interference in Full Duplex setting for **improving spectral efficiency** of **5G systems** by **2x**
- Modelled a realistic communication channel and developed an application for simulation in Tkinter
- Achieved more than **120dB** SIC[7] & up to **20dB** SINR[8]; linearized PA[9] using Indirect Learning Approach

Awards & Achievements

- National Prize Certificate holder for top students at IIT Madras (Amongst top **7%** entrants) *2017*
- Secured AIR[10] **323** in JEE Advanced (top **0.16%**) & AIR **491** in JEE Mains (top **0.03%**) *2016*
- Ranked in top **1%** in country in Stage I of National Science Olympiad (Chem.) conducted by HBCSE[11] *2015*
- Selected for KVPY[12] Fellowship, ranked **621st** in all over India (Amongst top **0.6%**) *2015*
- Awarded NTSE[13] Scholarship (Stage II - AIR **516**, Stage I – Maharashtra **state topper**) *2012*

Technical Skills

Deep Learning	*PyTorch, Tensorflow, GCloud, AWS, Adobe Sensei*
Programming Languages	*Python, C/C++, MATLAB, Java, JavaScript, FORTRAN, Verilog*
Software Tools	*Git, PyCharm, LaTeX, MS-Office*

Mentorship & Volunteering Experience

Mentor at Media and Data Science Research Lab, Adobe, Noida *May – Aug 2021*

- Mentored **4** research interns working on various problems in the fields of Reinforcement Learning & Document Structure Extraction

Moderator at Liberty, Equality and Fraternity (LEAF) Society, IIT Madras *Jan 2021 – May 2021*

- LEAF, a society of **20+ students** established with intention of **instilling culture of healthy discussions** on diverse **socio-economic issues** concerning the world, and India in particular
- As an **elected** moderator, conducted **8** sessions over the period of **4** months discussing issues like *'Indian Agrarian Distress', 'Global Human Rights Situation', 'Poverty in India', 'Economic Impact of COVID-19',* etc.
- Recorded proceedings of all sessions, ensured peaceful resolutions of debates, conducted polls for topics

Coordinator at Extra Mural Lectures (EML), IIT Madras *Jun 2017 – May 2018*

- Worked in a team of **15** to organize **20** lectures of distinguished dignitaries throughout the year
- Co-ordinated **event publicity** through posters, trailers, bios of speakers on the campus & social media
- Organized **a lecture in SRM** University as a part of EML-Omnibus, an initiative to expand EML **outside of IITM**

[5] *National Supercomputing Mission*
[6] *Numerical Linear Algebra*
[7] *Self-Interference Cancellation*
[8] *Signal to Interference + Noise Ratio*
[9] *Power Amplifier*
[10] *All India Rank*
[11] *Homi Bhabha Centre for Science Education*
[12] *Kishore Vaigyanik Protsahan Yojana*
[13] *National Talent Search Examination*